

Deploying and Improving AI/ML based Fault Classification and Fault Location Detection Algorithms of Transmission Lines on Edge Devices

Sai Rishik Thakkallapally

20MCME05
University of Hyderabad

Abstract

The detection, classification, and localization of faults in long transmission lines are critical for ensuring the reliability and stability of power systems. Transmission lines are prone to faults caused by environmental factors, equipment malfunctions, and other unforeseen events, which can lead to significant disruptions and, in extreme cases, system-wide failures. Timely and accurate monitoring of such faults is essential to maintain the safe and efficient operation of the power grid.

While traditional physics-based methods have been widely utilized for fault detection, classification, and localization, their limitations in handling complex and dynamic power systems present challenges in ensuring optimal performance. Advances in artificial intelligence and machine learning (AI/ML) offer promising solutions to address these limitations by enhancing the precision, speed, and adaptability of fault management techniques. AI/ML-based models can analyze large datasets, detect intricate patterns, and provide real-time insights, effectively complementing traditional approaches.

This research aims to bridge the gap between traditional physics-based fault management techniques and modern AI/ML-driven methods. By leveraging AI/ML models for fault classification and localization, optimizing them for lightweight deployment and validating their performance on real-world systems, this study seeks to enhance the reliability and efficiency of fault detection systems. The ultimate goal is to develop a robust fault management framework that ensures the stability and security of power systems while paving the way for smarter and more resilient energy infrastructure.

Internal Supervisor:

Name - Dr Mridula Verma
Institute - University of Hyderabad
Signature - _____

External Supervisor:

Name - Sudhanshu Mishra
Institute - GE Vernova (General Electric)
Signature - 

References

1. Kunjin Chen, Caowei Huang, Jinliang He , et al. "Fault detection, classification and location for transmission lines and distribution systems: a review on the methods ", pp. 1-9, 2016
2. Rui Fan, Tianzhixi Yin, Renke Huang, Jianming Lian,, and Shaobu Wang, , et al. "Transmission Line Fault Location Using Deep Learning Techniques ", pp. 1-5, 2023